MATH2014

Solution to Question Bank

Chapter 1

Q1

The function is **NOT** continuous at $(0,0)$ as

$$\lim_{(x,y)\to(0,0)} \frac{\sqrt{xy+1}-1}{xy} = \lim_{(x,y)\to(0,0)} \frac{1}{\sqrt{xy+1}+1} = \frac{1}{2} \neq 0$$

Q2

The function is **NOT** continuous at $(0,0)$ as

$\lim_{x\to 0, y=x^2} f(x,y) = \lim_{x\to 0} \frac{2x^4}{x^4+x^4} = 1$, and

$\lim_{x\to 0, y=-x^2} f(x,y) = \lim_{x\to 0} \frac{-2x^4}{x^4+x^4} = -1$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

Q3

Consider the critical point of f on R and find the maximum and minimum values of f on the boundary of R.

Note that $f = (x^2+y^2-2x+2)^{\frac{1}{2}}$.

Then, $\begin{cases} f_x = \frac{1}{2}(x^2+y^2-2x+2)^{-1/2}(2x-2) = 0 \\ f_y = \frac{1}{2}(x^2+y^2-2x+2)^{-1/2}2y = 0 = 0 \end{cases} \implies (x,y) = (1,0)$

With the constraint $x^2+y^2 = 4, y \geq 0, x \in [-2,2], f(x,y) = \sqrt{6-2x}$, with maximum $\sqrt{10}$ and minimum $\sqrt{2}$.

When $y = 0$, $f(x,y) = \sqrt{x^2-2x+2}$ has maximum $\sqrt{10}$ at $(-2,0)$ and minimum 1 at $(1,0)$.

Hence, the absolute maximum $\sqrt{10}$ and minimum 1.

Q4

Note that $\begin{cases} f_x = 2xy(4-x-y) - x^2y = -3x^2y + 8xy - 2xy^2 = 0 \\ f_y = x^2(4-x-y) - x^2y = -x^3 + 4x^2 - 2x^2y = 0 \end{cases}$

When $x = 0$ or $y = 0, f = 0$.

Now consider $x \neq 0$ and $y \neq 0$, $\begin{cases} 3x+2y = 8 \\ x+2y = 4 \end{cases} \implies (2,1)$ is a critical point.

Consider the boundary $x+y = 6, f(x) = -2x^2(6-x)$.

$f'(x) = 0 \implies x = 4, (4,2)$ is a local minimum as $f''(s) > 0$.

Hence, $f(4,2) = -64$ is the absolute minimum and $f(2,1) = 4$ is the absolute maximum.

Q5 Note that $T = 20+2x+2y+z^2, g = x^2+y^2+z^2-11 = 0, h = x+y+z-3 = 0$.

Now, $\nabla T - \lambda\nabla g - \mu\nabla h = 0$

$$\begin{cases} 2 = 2x\lambda + \mu \\ 2 = 2y\lambda + \mu \\ 2z = 2z\lambda + \mu \\ x^2 + y^2 + z^2 = 11 \\ x + y + z = 3 \end{cases} \implies \lambda(x - y) = 0$$

i) $\lambda = 0 \implies (x, y, z) = (3, -1, 1)$ or $(-1, 3, 1)$.
ii) $\lambda \neq 0 \implies x = y, (x, y, z) = (1 + \frac{2}{3}\sqrt{3}, 1 + \frac{2}{3}\sqrt{3}, 1 - \frac{4}{3}\sqrt{3})$ or $(1 - \frac{2}{3}\sqrt{3}, 1 - \frac{2}{3}\sqrt{3}, 1 + \frac{4}{3}\sqrt{3})$
Compare the four critical points,
$T(3, 1, -1) = T(-1, 3, 1) = 25$ and
$T(1 + \frac{2}{3}\sqrt{3}, 1 + \frac{2}{3}\sqrt{3}, 1 - \frac{4}{3}\sqrt{3}) = T(1 - \frac{2}{3}\sqrt{3}, 1 - \frac{2}{3}\sqrt{3}, 1 + \frac{4}{3}\sqrt{3}) = \frac{91}{3}$.
Hence, the minimum is $T = 25$ and the maximum is $T = \frac{91}{3}$

Q6

$f = x^2 + 4y^2 - 9 = 0, g = 18y - 14x^2 + 45 = 0.$

$x_1 = 1 + h, y_1 = -1 + k.$

Now,

$f(1, -1) = -4,\ f_x(1, -1) = 2x\Big|_{x=1} = 2,\ f_y(1, -1) = 8y\Big|_{y=-1} = -8$

$g(1, -1) = 13,\ g_x(1, -1) = -28x\Big|_{x=1} = -28,\ g_y(1, -1) = 18.$

$$\begin{cases} 2h - 8k = 4 \\ -28h + 18k = -13 \end{cases} \implies \begin{cases} h = 0.1702 \\ k = -43/94 = -0.457 \end{cases} \implies \begin{cases} x \approx 1.17 \\ y \approx -1.46 \end{cases}$$

Q7

Note that $f_x = 3x^2 - 12.y_y = 3y^2 - 3, f_{xx} = 6x,\ f_{yy} = 6y$ and $f_{xy} = f_{yx} = 0.$

$$\begin{cases} f_x = 0 \implies x = \pm 2 \\ f_y = 0 \implies y = \pm 1 \end{cases} \implies (x, y) = (2, 1), (2, -1), (2-, 1), \text{ or (-2,-1).}$$

At $(2, 1)$, $H = 12 \cdot 6 - 0 = 72 > 0, A = 12 > 0$

At $(2, -1)$, $H = 12 \cdot -6 = -72, A = 12 > 0$

At $(2, -1)$, $H = -12 \cdot -6 = -72, A = -12 < 0$

At $(-2, -1)$, $H = -12 \cdot -6 = 72, A = -12 < 0$

Hence $(2, -1)$ and $(-2, 1)$ are saddle points, $(2, 1)$ is a local minimum and $(-2, -1)$ is a local maximum.

Q8 Note that $f_x = -2x + 2,\ f_y = -2y + 4,\ f_{xx} = -2,$ and $f_{xy} = f_{yx} = 0.$

$$\begin{cases} f_x = 0 \\ f_y = 0 \end{cases} \implies \begin{cases} x = 1 \\ y = 2 \end{cases}$$

Now, $H = f_{xx}(1, 2)f_{yy}(1, 2) - [f_{xy}(1, 2)]^2 = 4 > 0$, and $A = f_{xx} = -2 < 0$.

Thus, $f(x, y)$ has a maximum at $(1, 2)$ and $f(1, 2) = 7$.

On the boundary, when $x = 0$, the maximum of $-y^2 + 4y + 2$ is $y = 2$, $f(0, 2) = 6$;
when $y = 0$, the maximum of $-x^2 + 2x + 2$ is at $x = 1$, $f(1, 0) = 3$.
Thus, there is one absolute maximum at $(1, 2)$ and $f(1, 2) = 7$.
When x or y tends to infinity, $f(x, y)$ tends to negative infinity. Thus, there is no absolute minima.

Q9

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt} = (2xe^{2y}\cos 3z)(-\sin t) + (2x^2e^{2y}\cos 3z)(\frac{1}{t+2}) + (-3x^2e^{2y}\sin 3z)\cdot 1$$
$$= -2xe^{2y}\cos 3z \sin t + \frac{2x^2e^{2y}\cos 3z}{t+2} - 3x^2e^{2y}\sin 3z; \text{ at the point on curve } z = 0 \implies t = z = 0$$
$$\implies \left.\frac{dw}{dt}\right|_{(1,\ln 2,0)} = 0 + \frac{2\cdot 1\cdot 4\cdot 1}{2} - 0 = 4.$$

Q10

Solving the equations $f_x = y + 2 - \frac{2}{x} = 0$ and $f_y = x - \frac{1}{y} = 0$, we obtain $(\frac{1}{2}, 2)$ is the only critical point of the function $f(x, y) - xy + 2x - \ln x^2y$. Since $f_{xx} = \frac{2}{x^2}, f_{yy} = \frac{1}{y^2}$ and $f_{xy} = 1$, we have $f_{xx}(\frac{1}{2}, 2)f_{yy}(\frac{1}{2}, 2) - [f_{xy}(\frac{1}{2}, 2)]^2 = 8 \times \frac{1}{4} - 1 = 1 > 0$. Thus, the point is a relative minimum.

Q11

Note that $f_x = 6xy - 6x, f_y = 3x^2 + 3y^2 - 6y, f_{xx} = 6y - 6, f_{yy} = 6y - 6, f_{xy} = 6x$.
We have $(0, 0)$ is a relative maximum, $(0, 2)$ is a relative minimum, $(1, 1)$ and $(-1, 1)$ are saddle points.

Q12

Set $x = r\cos\theta, y = r\sin\theta$.
Then, $f(x, y) = \frac{x^2y}{x^2 + y^2} = r\sin\theta\cos^2\theta$.

Now, $\lim_{(x,y)\to(0,0)} f(x, y) = \lim_{r\to 0} r\sin\theta\cos^2\theta = 0 = f(0, 0)$ $(|\sin\theta\cos^2\theta| \le 1)$. Hence, f is continuous at $(0, 0)$.

Q13

Let's do the interior critical point fist. As usual, we need to find points when both partial derivatives of f are 0. Computing gives us:

$$\begin{cases} f_x(x, y) = 4x^3 - 4y \\ f_y(x, y) = 4y^3 - 4x \end{cases}$$

Then, we can find the critical points are $(0, 0), (1, 1), (-1, -1)$.
But notice the our region is defined by $0 \le x \le 3$ and $0 \le y \le 2$. The point $(-1, -1)$ is not in the domain. So forget about it. We get two critical points that we care about: $(0, 0)$ and $(1, 1)$.
Next, we have to check along the boundaries. We can describe the boundary curves by making all the inequalities equalities in the description of D. This gives us four curves $x = 0, x = 3, y = 0, y = 3$.
We consider both maximum and minimum points and points on these boundaries:
Lets start with $x = 0$; when $x = 0$, the function f becomes:

$f(y) = f(0, y) = 0^4 + y^4 - 4 \cdot 0 + 2 = y^4 + 2$,
then, **we get** $(0, 0)$ **and** $(0, 2)$.
Next, we will look at the boundary curve $x = 3$. On this curve the function turns into $f(y) = f(3, y) = 3^4 + y^4 - 4 \cdot 3 \cdot y + 2 = y^4 - 12y + 83$.
To find any points on this boundary, take the derivative and set it equal to 0:
$0 = f'(y) = 4y^3 - 12$.
So, $y = \sqrt[3]{3}$ which gives us the point $(3, \sqrt[3]{3})$ (which is in our domain, since $\sqrt[3]{3}$ between 0 and 2).
Again, we also have to check at the ends of the intervals $0 \leq y \leq 2$. When $y = 0$ and $y = 2$, we get the points $(3, 0)$ and $(3, 2)$.
Next, we'll check $y = 0$. The function f becomes:
$f(x) = f(x, 0) = x^4 + 0^4 - 4x(0) + 2 = x^4 + 2$.
Taking the derivative and setting it equal to 0 gives us $4x^3 = 0 \implies x = 0$.
No new point we get from this case.
The last boundary curve to check is $y = 2$. Then
$f(x) = f(x, 2) = x^4 - 8x + 2$.
Setting the derivative equal to zero yields, $4x^3 - 8 = 0 \implies x = \sqrt[3]{2}$. This gives us the point $(\sqrt[3]{2}, 2)$, which is in our domain. Checking when $x = 0$ and $x = 3$ gives us points we already have.
Thus, our final list is:
$(0, 0), (1, 1), (0, 2), (3, \sqrt[3]{3}), (3, 0), (3, 2), (\sqrt[3]{2}, 2)$.
All we have to do now is plug all of these values into the function f and pick out the smallest and biggest:

$$
\begin{aligned}
f(0, 0) &= 2 \\
f(1, 1) &= 0 \\
f(0, 2) &= 18 \\
f(3, \sqrt[3]{3}) &= 84 - 9\sqrt[3]{3} \approx 70 \\
f(3, 0) &= 83 \\
f(3, 2) &= 75 \\
f(\sqrt[3]{2}, 2) &= 18 - 6\sqrt[3]{2} \approx 10.5.
\end{aligned}
$$

Hence, the minimum is 0 and the maximum is 83.

Q14

Note that, by the method of Lagrange multiplier, we have a system of equations:

$$
\begin{aligned}
yz &= \lambda 2x \\
xz &= \lambda 4y \\
xy &= \lambda 6z \\
x^2 + 2y^2 + 3z^2 &= 6
\end{aligned}
$$

Observe the following: we know that x, y, and z cannot all be 0. Otherwise, out constraint equation would give us $0 = 6$, which is false. We now consider all the other possibilities.

First, we'll consider the possibilities where only one of the variables is nonzero,
we get $(\pm\sqrt{6}, 0, 0), (0, \pm\sqrt{3}, 0), (0, 0, \pm\sqrt{2})$.
Next, we consider the case when two of the variables are nonzero and one of them is zero, we can conclude that it is impossible.
All that's left is to consider the case when all three are nonzero, we get $(\pm\sqrt{2}, \pm 1, \pm\sqrt{\frac{2}{3}})$.
As $f(\pm\sqrt{2}, \pm 1, \pm\sqrt{\frac{2}{3}}) = \pm\sqrt{2}(1)\sqrt{\frac{2}{3}} = \pm\frac{2}{\sqrt{3}}$.
And at all the other points, f are all 0.
So, maximum value of f on the surface $x^2 + y^2 + z^2 = 6$ is $\frac{2}{\sqrt{3}}$, and the minimum value is $-\frac{2}{\sqrt{3}}$.

Q15

We need to solve the equations:

$$(*) \begin{cases} a(x,y) = 0 & \text{where } a(x,y) = y - x^2 \\ b(x,y) = 0 & \text{where } b(x,y) = x^2 + y^2 = 1 \end{cases}.$$

We pick $(x_0, y_0) = (1, 1)$ as an approximate solution. If $a_0(x, y)$ and $b_0(x, y)$ denote the linear approximations at (x_0, y_0), then we obtain $a_0(x, y) = -2x + y + 1$ and $b_0(x, y) = 2x + 2y - 3$.
Hence, the equations $a_0(x, y) = 0$ and $b_0(x, y) = 0$ have the solution $(x_1, y_1) = (\frac{5}{6}, \frac{2}{3})$.

Q16

We calculate $\begin{cases} a(x,y) = \frac{\partial f}{\partial x}(x,y) = 4xy^2 + 2xy - 2 \\ b(x,y) = \frac{\partial f}{\partial y}(x,y) = 4x^2y + x^2 - 2y \end{cases}$

The critical points of $f(x, y)$ are the solutions to the equations $a(x, y) = 0$ and $b(x, y) = 0$.
We start with approximate solution $(x_0, y_0) = (1, -1)$ and we obtain the equations
$0 = a_0(x, y) = 0 + 2(x - 1) - 6(y + 1) = 2x - 6y - 8$.
$0 = b_0(x, y) = -1 - 6(x - 1) + 2(y + 1) = -6x + 2y + 7$.
The solution to these equations is the next approximate solution $(x_1, y_1) = (13/16, -17/16) = (0.8125, -1.0625)$.

Q17

We minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraints $0 = g(x, y, z) = x^2 + y^2 - z^2, 0 = h(x, y, z) = x - 2z - 3$.
To do so, we define the zuxiliary function
$L(x, y, z, \lambda, \mu) - f(x, y, z) + \lambda g(x, y, z) + \mu h(x, y, z) = x^2 + y^2 + z^2 + \lambda[x^2 + y^2 - z^2] + \mu[x - 2z - 3]$
and solve the system of equations:

(1) $0 = L_x(x, y, z, \lambda, \mu) = 2x + 2\lambda x + \mu = 2(1 + \lambda)x + \mu$

(2) $0 = L_y(x, y, z, \lambda, \mu) = 2y + 2\lambda y = 2(1 + \lambda)y$

(3) $0 = L_z(x, y, z, \lambda, \mu) = 2z - 2\lambda z - 2\mu = 2(1 - \lambda)z - 2\mu$

(4) $0 = L_\lambda(x, y, z, \lambda, \mu) = x^2 + y^2 - z^2$

(5) $0 = L_\mu(x, y, z, \lambda, \mu) = x - 2z - 3$

Then, we have two solution, $(x, y, z) = (-3, 0, -3), (1, 0, -1)$.
As $f(-3, 0, -3) = 3\sqrt{2}$, and $f(1, 0, -1) = \sqrt{2}$, the minimum is $\sqrt{2}$.

Q18

We minimize $f(x, y, z) = (x + 1)^2 + (y - 2)^2 + (z - 3)^2$, subject to
$$\begin{cases} g(x, y, z) = x + 2y - 3z - 4 = 0 \\ h(x, y, z) = 2x - y + 2z - 5 = 0 \end{cases}.$$

By the Lagrange Multiplier method, we solve the equations

$$\begin{cases} 2(x + 1) = \lambda + 2\mu \\ 2(y - 2) = 2\lambda - \mu \\ 2(z - 3) = -3\lambda + 2\mu \\ x + 2y - 3z = 4 \\ 2x - y + 2z = 5 \end{cases}$$

to obtain $(x, y, z) = (\frac{37}{15}, \frac{49}{15}, \frac{25}{15})$.
Physical consideration indicates that this must be a minimum. Therefore, the shortest distance is given by
$\sqrt{(\frac{37}{15} + 1)^2 + (\frac{49}{15} - 2)^2 + (\frac{25}{15} - 3)^2} = \sqrt{\frac{77}{5}} = 3.9243.$

Q19 Use Lagrange multipliers method:

$L(h, s, \lambda) = 200h^{2/3}s^{1/3} - \lambda(20h + 170s - 20,000)$.
$$\begin{cases} 0 = 200 \cdot \frac{2}{3}h^{-1/3}s^{1/3} - 20\lambda \\ 0 = 200 \cdot \frac{1}{3}h^{2/3}s^{-2/3} - 170\lambda \\ 20h + 170s = 20,000 \end{cases} \implies \begin{cases} h = \frac{2,000}{3} \approx 666.667 \\ s = \frac{2,000}{51} \approx 39.2157 \\ \lambda = \sqrt[3]{\frac{8,000}{459}} \approx 2.593 \end{cases}$$

Hence, $R(h, s) = 200 \times (2000/3)^{2/3}(2000/51)^{1/3} \approx 51854.82 \approx 51855$.

Q20

Let $S = \frac{x}{x - y}$, then $\frac{\partial S}{\partial x} = \frac{-y}{(x - y)^2}$ and $\frac{\partial S}{\partial y} = \frac{x}{(x - y)^2}$.
Note that $|\Delta| \approx |dS| \leq |S_x(9, 5)||\Delta x| + |S_y(9, 5)||\Delta y|$
$= \left|\frac{-5}{(9 - 5)^2}\right||0.05| + \left|\frac{9}{(9 - 5)^2}\right||0.1| = 0.0718575 \leq 0.072.$

Q21

Applying Newton's method to solve the equations $f_x = 0$ and $f_y = 0$ yields the matrix equation

$$\begin{bmatrix} -2y_0 & -2x_0 \\ -2x_0 & 2 \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} = - \begin{bmatrix} -2x_0y_0 \\ 2 - x_0^2 + 2y_0 \end{bmatrix}.$$

At the current point $(x_0, y_0) = (-3, -3)$, we have the matrix equation

$$\begin{bmatrix} 6 & 6 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} x+3 \\ y+3 \end{bmatrix} = \begin{bmatrix} 18 \\ 13 \end{bmatrix}.$$

Solving the equations yields

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1.25 \\ -1.75 \end{bmatrix}$$

Q22

Note that $\begin{cases} f(x,y) = x^2 + y^2 - 4, \quad f_x(x,y) = 2x, \quad f_y(x,y) = 2y \\ g(x,y) = e^x + y - 1, \quad g_x(x,y) = e^x, \quad g_y(x,y) = 1 \end{cases}$.

First, iteration: $(x_0, y_0) = (1, -1.7)$

$$D(1,-1.7) = \begin{vmatrix} f_x(1,-1.7) & f_y(1,-1.7) \\ g_x(1,-1.7) & g_y(1,-1.7) \end{vmatrix} = 11.24216$$

$$D_x(1,-1.7) = \begin{vmatrix} -f(1,-1.7) & f_y(1,-1.7) \\ -g(1,-1.7) & g_y(1,-1.7) \end{vmatrix} = 4.784179 \times 10^{-2}$$

$$D_y(1,-1.7) = \begin{vmatrix} f_x(1,-1.7) & -f(1,-1.7) \\ g_x(1,-1.7) & -g(1,-1.7) \end{vmatrix} = -0.3355747$$

$$\implies \begin{cases} x_1 = 1 + \dfrac{4.784179 \times 10^{-2}}{11.24216} = 1.0043 \\ y_1 = -1.7 + \dfrac{-0.3355747}{11.24216} = -1.7299 \end{cases}$$

Q23

Note that

$$\begin{aligned} f(x,y) &= x^2 + x - y^2 - x \\ g(x,y) &= y - \sin(x^2) \\ f_x(x,y) &= 2x + 1 \\ f_y(x,y) &= -2y \\ g_x(x,y) &= -2\cos(x^2)x \\ g_y(x,y) &= 1 \end{aligned}$$

First iteration: $(x_0, y_0) = (0.7, 0.5)$.

$$D(x,y) = \begin{vmatrix} f_x(x,y) & f_y(x,y) \\ g_x(x,y) & g_y(x,y) \end{vmatrix}$$

$$D(0.7,0.5) = \begin{vmatrix} f_x(0.7,0.5) & f_y(0.7,0.5) \\ g_x(0.7,0.5) & g_y(0.7,0.5) \end{vmatrix} = 1.164733998$$

$$D_x = \begin{vmatrix} -f(x,y) & f_y(x,y) \\ -g(x,y) & g_y(x,y) \end{vmatrix}$$

$$D_x(0.7, 0.5) = \begin{vmatrix} -f(0.7,0.5) & f_y(0.7,0.5) \\ -g(0.7,0.5) & g_y(0.7,0.5) \end{vmatrix} = 3.062588817 \times 10^{-2}$$

$$D_y = \begin{vmatrix} f_x(x,y) & -f(x,y) \\ g_x(x,y) & -g(x,y) \end{vmatrix}$$

$$D_y(0.7, 0.5) = \begin{vmatrix} f_x(0.7,0.5) & -f(0.7,0.5) \\ g_x(0.7,0.5) & -g(0.7,0.5) \end{vmatrix} = 3.618091734 \times 10^{-3}$$

$$\implies \begin{cases} x_1 = 0.7 + \dfrac{3.062588817 \times 10^{-2}}{1.164733998} = 0.7262943198 \\ y_1 = 0.5 + \dfrac{3.618091734 \times 10^{-3}}{1.164733998} = 0.5031062764 \end{cases}$$

Q24

$$\begin{cases} f_x = x^x(1 + \ln x)\sin y = 0 := p(x,y) \\ f_y = x^x \cos y = 0 := g(x,y) \end{cases}$$

Let $x_1 = 0.5 + h, y_1 = 1.5 + k$.

Then,

$p(0.5, 1.5) = 0.21821, g(0.5, 1.5) = 0.05.$

$p_x = x^x(1 + \ln x)^2 \sin x + x^{x-1} \sin y$

$p_x(0.5, 1.5) = 1.47708$

$p_y = x^x(1 + \ln x)\cos y, p_y(0.5, 1.5) = 0.01535$

$g_x = x^x(1 + \ln x)\cos x, g_x(0.5, 1.5) = 0.01535$

$g_y = -x^2 \sin y, g_y(0.5, 1.5) = -0.70534$

Now, $\begin{cases} 1.47708h + 0.015k = -0.21821 \\ 0.01535h - 0.70534l = -0.05 \end{cases} \implies \begin{cases} h = -0.14843 \\ k = 0.06766 \end{cases}$

Hence, $x_1 = 0.3516, y_1 = 1.5677$.

Q25 Find the maximum and minimum of $f(x, y, z) = x^2 + y^2 + z^2$

with constraints $\begin{cases} x + y + z = 0 \\ x^2 + y^2 + 4z^2 = 1 \end{cases}$.

Now, $L(x, y, z, \lambda, \mu) = x^2 + y^2 + z^2 - \lambda(x + y + z) - \mu(x^2 + y^2 + 4z^2 - 1)$

$$\implies \begin{cases} L_x = 2(1-\mu)x - \lambda = 0 \quad (1) \\ L_y = 2(1-\mu)y - \lambda = 0 \quad (2) \\ L_z = 2(1-4\mu)z - \lambda = 0 \quad (3) \\ x + y + z = 0 \\ x^2 + y^2 + 4z^2 = 1 \end{cases}$$

Now, $(1) \times (1 - 4\mu) + (2) \times (1 - 4\mu) + (3) \times (1 - \mu) \implies 3\lambda(1 - 3\mu) = 0$

$\implies \lambda = 0$ or $\mu = 1/3$.

Case 1 $\lambda = 0$.

$(1) + (2) + (3) \implies 6\mu z = 0$.

As $\mu \neq 0$, then $z = 0$.

$\implies (x, y, z) = (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0) \implies f = 1$

Case 2 $\mu = 1/3$.

From $(1), (2), (3), x = \frac{3}{4}\lambda, y = \frac{3}{4}\lambda, z = \frac{-3}{2}\lambda$

$\implies (x, y, z) = (\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{6}, -\frac{\sqrt{2}}{3})$, or $(-\frac{\sqrt{2}}{6}, -\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{3})$

$\implies f = 1/3$

Thus, $a = 1, b = \frac{1}{\sqrt{3}}$ and area $= \pi ab = \frac{\pi}{\sqrt{3}}$.

Q26 We apply Lagrange Multiplier Method to the function $T = 2x^2 + 5y^2 + 11z^2 + 2xy - 4xz + 16yz$, subject to $g = x^2 + y^2 + z^2 - 1 = 0$, to obtain the equations

$$\begin{cases} 4x + 20y - 4z = \lambda 2x \\ 20x + 10y + 16z = \lambda 2y \\ -4x + 16y + 22z = \lambda 2z \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

The first three equations are equivalent to an eigenvalue problem

$$\begin{bmatrix} 4 & 20 & -4 \\ 20 & 10 & 16 \\ -4 & 16 & 22 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Solving the characteristic equation $\begin{vmatrix} 4-\lambda & 20 & -4 \\ 20 & 10-\lambda & 16 \\ -4 & 16 & 22-\lambda \end{vmatrix} = 0$, one obtains eigenvalues $\lambda = 9, -9$, and 18.

Corresponding eigenvectors are obtained as follows: when $\lambda = -9$, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ -2t \\ 1 \end{bmatrix}$; when $\lambda = 9$, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2s \\ s \\ -2s \end{bmatrix}$;

when$\lambda = 18$, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \\ 2r \\ 2r \end{bmatrix}$.

We then substitute these solution into the constraint and obtain $t = \pm 1/3, s = \pm 1/3, r = \pm 1/3$.

Therefore, the critical points of the function T on the sphere are $(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}), (-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}), (\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$, $(-\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}), (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$ and $(-\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3})$.

Substituting these points into T, we conclude that $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$ and $(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$ are the hottest points on the sphere with maximum temperature $= 18$.

Q27

We have $\dfrac{\partial z}{\partial x} = f' \cos x$ and $\dfrac{\partial z}{\partial y} = \cos y - f' \cos y$.
Thus, $\sec x f' \cos x + \sec y(\cos y - f' \cos y) = f' + 1 - f' = 1$.

Q28

Not continuous.

Proof: Let $y = x$,

then $\lim\limits_{x\to 0} f(x,y) = \lim\limits_{x\to 0} \dfrac{x^4}{x^4} = 1 \neq 0$
Thus, the function $f(x,y)$ is not continuous at $(0,0)$.

Q29

$f_x = (-2x)(y - x^4) + (y - x^2)(-4x^3) = 2x(3x^4 - 2x^2y - y) = 0$ ①

$f_y = (y - x^4) + (y - x^2) = 2y - x^2 - x^4 = 0$ ②

From ① $\Rightarrow x = 0$ or $3x^4 - 2x^2y - y = 0$ ③ $\Rightarrow$ critical point $(0,0)$

From ② $\Rightarrow y = \frac{x^2+x^4}{2}$ ④.

From ③, ④ $\Rightarrow 2x^4 - 3x^2 + 1 = 0$

$x = \pm 1$ or $\pm\frac{\sqrt{2}}{2} \Rightarrow$ critical points $(1,1), (-1,1), (\frac{\sqrt{2}}{2}, \frac{3}{8}), (-\frac{\sqrt{2}}{2}, \frac{3}{8})$

$A = f_{xx} = 30x^4 - 12yx^2 - 2y$

$B = f_{xy} = -2x - 4x^3$

$C = f_{yy} = 2$, $H = AC - B^2$

critical point	A	B	C	H	nature
(0,0)	0	0	2	$= 0$	inconclusive
(1,1)	16	- 6	2	< 0	saddle point
(-1,1)	16	6	2	< 0	saddle point
$(\frac{\sqrt{2}}{2}, \frac{3}{8})$	$\frac{9}{2}$	$-2\sqrt{2}$	2	> 0	local min
$(-\frac{\sqrt{2}}{2}, \frac{3}{8})$	$\frac{9}{2}$	$2\sqrt{2}$	2	> 0	local min

Q30

Let $(x^2 - y^2) - \lambda(x^2 + 2y^2 + 3z^2) = 0$

$$\begin{cases} 2x - 2\lambda x = 0 ① \\ -2y - 4\lambda y = 0 ② \\ 0 = 6\lambda z ③ \\ x^2 + 2y^2 + 3z^2 = 1 ④ \end{cases}$$

case 1 $\lambda = x = y = 0 \Rightarrow z = \pm\frac{\sqrt{3}}{3}$ Thus, critical point $(0, 0, \pm\frac{\sqrt{3}}{3})$

case 2 $\lambda \neq 0$ from ③ $\Rightarrow z = 0$

If $x = 0 \Rightarrow y = \pm\frac{\sqrt{2}}{2}$ Thus, critical point $(0, \pm\frac{\sqrt{2}}{2}, 0)$

If $x \neq 0$, from ①$\Rightarrow \lambda = 1$ from ② $\Rightarrow y = 0$ Thus, $x = \pm 1$, critical points $(\pm 1, 0, 0)$

$f(0, 0, \pm\frac{\sqrt{3}}{3}) = 0$

$f(0, \pm\frac{\sqrt{2}}{2}, 0) = -\frac{1}{2}$ (min)

$f(\pm 1, 0, 0) = 1$ (max)

Q31

Not continuous.

Let $y = \sqrt{x}$, then $\lim_{(x,y)\to(0,0)} f(x,y) = \frac{1}{8} \neq 0$.

Thus, the function is not continuous at $(0,0)$.

Q32

Let $u = x + 2y, v = 2y - 3x, z = u^v$,

then $\dfrac{\partial z}{\partial x} = \dfrac{\partial z}{\partial u}\dfrac{\partial u}{\partial x} + \dfrac{\partial z}{\partial v}\dfrac{\partial v}{\partial x}$

$= vu^{v-1} + u^v \ln u(-3)$

$= (2y - 3x)(x + 2y)^{(2y-3x-1)} - 3(x + 2y)^{(2y-3x)} \ln(x + 2y)$

Chapter 2

Q33

The points of intersection are $1 + \sin\theta = 3\sin\theta \implies \sin = 1/2, \theta = \pi/6, 5\pi/6$.

Thus, Area $= 2 \times \left(\int_{-\pi/2}^{\pi/6} \int_0^{1+\sin\theta} r dr d\theta - \int_0^{\pi/6} \int_0^{3\sin\theta} r dr d\theta \right)$

$= 2 \times [(\dfrac{\pi}{2} - \dfrac{9\sqrt{3}}{16}) - (\dfrac{3\pi}{8} - \dfrac{9\sqrt{3}}{16})] = 2 \times \dfrac{\pi}{8} = \dfrac{\pi}{4}$.

Q34

Let $u = x^2 - y^2, v = \dfrac{y}{x}$.

Now, $\dfrac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 2x & -2y \\ -y/x^2 & 1/x \end{vmatrix} = 2 - 2v^2$.

Thus, $\int\int_R \dfrac{y}{x} dx dy = \int\int_R \dfrac{v}{2(1-v^2)} du dv = \int_9^{1/2} \int_1^4 \dfrac{v}{2(1-v^2)} du dv$

$= \dfrac{-3}{4} \ln(1 - v^2)\Big|_0^{1/2} = \dfrac{-3}{4} \ln \dfrac{3}{4}$.

As there is one symmetric region in the third quadrant, the answer to this question is $\dfrac{-3}{2} \ln \dfrac{3}{4}$.

Q35 Note that $x^2 + y^2 + z^2 = a^2$.

a) rectangular coordinate : Set $\rho = k\sqrt{x^2 + y^2 + z^2}$.

Now, $m = \int\int\int p dw = k \int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} \sqrt{x^2 + y^2 + z^2} dz dy dx$

cylindrical coordinate: Set $\rho = k\sqrt{r^2 + z^2}$.

Now, $m = k \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} \sqrt{r^2 + z^2} r dz dr d\theta$

b) $m = k \int_0^{2\pi} \int_0^{\pi/2} \int_0^a \rho^3 \sin\phi d\rho d\phi d\theta$

c) $\dfrac{ka^4\pi}{2}$

Q36

Let $u = x - 2y, v = x + y$, then $x = \dfrac{u+2v}{3}, y = \dfrac{v-u}{3}$.

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} = 3.$$

Now, $x^2 - xy - 2y^2 = uv$.

Then $\displaystyle\int_0^1 \int_0^1 uv \cdot \frac{1}{3} du dv = \frac{1}{3}\int_0^1 \frac{1}{2}u^2 v\Big|_0^1 dv = \frac{1}{6}\int_0^1 v dv = \frac{1}{12}$.

Q37

$$4\int_0^{\pi/2}\int_0^{a\cos\theta}\int_0^{\sqrt{a^2-r^2}} r dz dr d\theta = 4\int_0^{\pi/2}\int_0^{a\cos\theta} r\sqrt{a^2-r^2} dr d\theta$$

$$= -2\int_0^{\pi/2} \frac{2}{3}(a^2-r^2)^{3/2}\Big|_0^{a\cos\theta} d\theta = 2\int_0^{\pi/2}(\frac{2}{3}a^3 - \frac{2}{3}a^3\sin^3\theta)d\theta$$

$$= \frac{2}{3}a^3\pi - \frac{4}{3}a^3(\frac{\cos^3\theta}{3} - \cos\theta)\Big|_0^{\pi/2} = \frac{2}{3}a^3\pi - \frac{4}{3}a^3\frac{3}{2} = \frac{2}{3}a^3(\pi - \frac{4}{3}).$$

Q38

$$\int_0^1\int_0^{1-x}\int_0^{\sqrt{1-x-y}} dz dy dx = \int_0^1\int_0^{1-x}\sqrt{1-x-y}dy$$

$$= \frac{2}{3}\int_0^1 (1-x)^{3/2}dx = \frac{4}{15}.$$

Q39

$$\int_0^1\int_{x^4}^{x^3}(x+y)dydx + \int_{-1}^0\int_{x^3}^{x^4}(x+y)dydx = \int_0^1\left[\frac{x^2}{2}+yx\right]_{x^4}^{x^3}dx + \int_{-1}^0\left[\frac{x^2}{2}+yx\right]_{x^3}^{x^4}dx$$

$$= \int_0^1(x^4 + \frac{x^6}{2} - x^5 - \frac{x^8}{2})dx + \int_{-1}^0(x^5 + \frac{x^8}{2} - x^4 - \frac{x^6}{2})dx$$

$$= \left[\frac{x^5}{5} + \frac{x^7}{14} - \frac{x^6}{6} - \frac{x^9}{18}\right]_0^1 + \left[\frac{x^6}{6} + \frac{x^9}{18} - \frac{x^5}{5} - \frac{x^7}{14}\right]_{-1}^0 = \frac{31}{630} - \frac{241}{630} = \frac{-1}{3}$$

Q40

Let $u = xy$ and $v = xy^3$, where $4 \le u \le 8$ and $5 \le v \le 15$.

Jocobian:

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial(xy)}{\partial x} & \frac{\partial(xy^3)}{\partial x} \\ \frac{\partial(xy)}{\partial y} & \frac{\partial(xy^3)}{\partial y} \end{vmatrix} = \begin{vmatrix} y & y^3 \\ x & 3xy^2 \end{vmatrix} = 2xy^3 = 2v.$$

Now, $\displaystyle\int\int_D dxdy = \int\int_{D_1}\frac{\partial(x,y)}{\partial(u,v)}dudv = \int\int_{D_1}\frac{1}{\frac{\partial(u,v)}{\partial(x,y)}}dudv = \int_4^8\int_5^{15}\frac{1}{2v}dvdu$

$$= \int_4^8\left[\frac{\ln v}{2}\right]_5^{15}du = \int_4^8(\frac{\ln 15}{2} - \frac{\ln 5}{2})du = \int_4^8(\frac{1}{2}\ln 3)du = 2\ln 3.$$

Q41

Set $u = x + y, v = x - y$, then $x = \dfrac{u+v}{2}, y = \dfrac{u-v}{2}$.

Note that $\dfrac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = -1/2$

Now, $\int\int_R (\frac{x-y}{x+y+2})^2 dxdy = \int\int_R (\frac{v}{u+2})\frac{1}{2}dudv = \int_{-1}^{1}\int_{-1}^{1}(\frac{v}{u+2})^2\frac{1}{2}dudv = \frac{2}{9}$.

Q42

In spherical coordinates: $x = \rho\sin\phi\cos\theta, y = \rho\sin\phi\theta, z = \rho\cos\phi$.

So the equation for the sphere in spherical coordinates is $\rho^2 = \rho\cos\phi \implies \rho = \cos\phi$.

So, $0 \le \theta \le 2\pi, 0 \le \phi \le \pi/4, 0 \le \rho \le \cos\phi$.

Then, the volume of the solid is

$V = \int\int\int_E 1dV = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{\cos\phi}\rho^2\sin\phi d\rho d\phi d\theta \int_0^{2\pi}\int_0^{\pi/4}\frac{1}{3}\rho^3\sin\phi\Big|_0^{\cos\phi}d\phi\theta$

$= \int_0^{2\pi}\int_0^{\pi/4}\frac{1}{3}\cos^3\phi\sin\phi d\phi d\theta = \int_0^{2\pi}\frac{-1}{12}\cos^4\phi\Big|_0^{\pi/4}d\theta = 2\pi(\frac{-1}{12}(\frac{1}{4}-1)) = \frac{\pi}{8}$.

Q43

From the integral, we know that $-1 \le x \le 1, 0 \le y \le \sqrt{1-x^2}, (x^2+y^2)^3 \le z \le 5$.

So for cylindrical coordinates: $x = r\cos\theta, y = r\sin\theta, z = z$.

$0 \le r \le 1, 0 \le \theta \le \pi, r^6 \le z \le 5$.

So $\int_{-1}^{1}\int_0^{\sqrt{1-x^2}}\int_{(x^2+y^2)^3}^{5}\sqrt{x^2+y^2}dzdydx = \int_0^1\int_0^{\pi}\int_{r^6}^{5} r\cdot rdzd\theta dr = \frac{14\pi}{9}$.

Q44

$A = \int_0^{\pi/4}\int_0^{\sin\theta} rdrd\theta + \int_{\pi/4}^{\pi/2}\int_0^{\cos\theta} rdrd\theta = \frac{1}{8}(\pi-2)$.

Q45

E is described by $x^2+z^2 \le y \le 4-x^2-z^2$ over a disk D in the xz-plane whose radius is given by the intersection of the two surface: $y = 4-x^2-z^2$ and $y = x^2+z^2$.

$4-x^2-z^2 = x^2+z^2 \implies x^2+z^2 = 2$. So the radius is $\sqrt{2}$.

Therefore, $V(E) = \int\int\int_E 1dV = \int\int_D\left[\int_{x^2+z^2}^{4-x^2-z^2} 1dy\right]dA = \int\int_D 4-2(x^2+z^2)dA$

$= \int_0^{2\pi}\int_0^{\sqrt{2}}(4-2r^2)rdrd\theta = \int_0^{2\pi}\left[2r^2-\frac{r^4}{2}\right]_0^{\sqrt{2}} = 4\pi$.

Q46

The top surface is $z = u_2(x,y) = \sqrt{9-x^2-y^2} = \sqrt{9-r^2}$ and the bottom surface is $z = u_1(x,y) = \sqrt{x^2+y^2} = r$ over the regoin D defined by the intersection of the top (or bottom) and the cylinder which is a disk $x^2+y^2 \le 1$ or $0 \le r \le 1$ in the xy-plane.

Now, $\int\int\int_E zdV = \int\int_D\left[\int_r^{\sqrt{9-r^2}} zdz\right]dA = \int_0^{2\pi}\int_0^1\int_r^{\sqrt{9-r^2}} zrdzdrd\theta$

$= \int_0^{2\pi}\int_0^1\frac{1}{2}(9-r^2-r^2)rdrd\theta = 4\pi$.

Q47

Here the region of integration is simple, but the function $f(x,y) = \cos(\frac{x-y}{x+y})$ is not. It seems reasonable to set $u = x-y, v = x+y$.

Then, as the point (x, y) varies in R, the point (u, v) varies in the triangular region Q bounded by the lines $v = u, v = 1, u = -v$.

By the change of variables formula

$\int\int_R \cos(\frac{x-y}{x+y})dxdy = \int\int_Q \cos(u/z)|J|dudv.$

Since $x = (u+v)/2, y = (u-v)/2$, we can compute the Jacobian factor: $J = x_u y_v - x_v y_u = 1/4 + 1/4 = 1/2$.

So, $\int\int_Q \cos(u/v)|J|dudv = \frac{1}{2}\int_0^1 \int_{-v}^{v} \cos(u/v)dudv = \frac{1}{2}\int_0^1 v\sin(u/v)\Big|_{-v}^{v} dv$

$= \sin 1 \int_0^1 vdv = \frac{\sin 1}{2}.$

Q48

Note that $\int_{-\pi/2}^{\pi/2}\int_{\cos\theta}^{2\cos\theta}\int_0^{3-r\sin\theta} rdzdrd\theta = \frac{9\pi}{4}.$

Q49

Where the hyperboloid intersects the cone, we have $r^2 + a^2 = 2r^2$, thus, $r = a$.

$V = \int_0^{2\pi} d\theta \int_0^a rdr \int_{\sqrt{2}r}^{\sqrt{a^2+r^2}} dz = \int_0^{2\pi} d\theta \int_0^a r(\sqrt{a^2+r^2} - \sqrt{2}r)dr$

$= \int_0^{2\pi}\left[\frac{1}{3}(a^2+r^2)^{3/2} - \frac{\sqrt{2}}{3}r^3\right]_0^a d\theta = \int_0^{2\pi}(\frac{1}{3}a^3\sqrt{2} - \frac{1}{3}a^3)d\theta$

$= \frac{2}{3}\pi a^3\sqrt{2} - \frac{2}{3}\pi a^3 = \frac{2}{3}\pi a^3(\sqrt{2}-1).$

Q50

Now, $\sigma = R - \rho$, the mass is

$M = \int\int\int \sigma dV = \int_0^{2\pi}\int_0^{\pi}\int_0^R (R-\rho)\rho^2\sin\phi d\rho d\phi d\theta$

$= \int_0^{2\pi}\int_0^{\pi}\left[(\sin\phi)(\frac{-1}{4}\rho^4 + \frac{1}{3}R\rho^3)\right]_0^R d\phi d\theta = \int_0^{2\pi}\int_0^{\pi}\frac{1}{12}R^4\sin\phi d\phi\theta$

$= \int_0^{2\pi}\frac{1}{6}R^4 d\theta = \frac{1}{3}\pi R^4.$

Q51

The solid's centroid $(\bar{x}, \bar{y}, \bar{z})$ lies on its axis of symmetry, here the z-axis. This makes $\bar{x} = \bar{y} = 0$. To find$\bar{z}$, we divide the first moment M_{xy} by the mass M.

To find the limits of integration for the mass and moment integrals, we continue with the four basic steps. We completed our initial sketch. The remaining steps give the limits of integration.

The $z - limits$. A line M through a typical point (r, θ) in the base parallel to the z-axis enters the solid at $z = 0$ and leaves at $z = r^2$.

The $r - limits$. A ray L through (r, θ) from the origin enters R at $r = 0$ and leaves at $r = 2$.

The $\theta - limits$. As L sweeps over the base like a clock hand, the angle θ it makes with the positive x-axis runs from $\theta = 0$ ro $\theta = 2\pi$. The value of M_{xy} is

$$M_{xy} = \int_0^{2\pi}\int_0^2\int_0^{r^2} zdz\, r\, drd\theta = \int_0^{2\pi}\int_0^2\left[\frac{z^2}{2}\right]_0^{r^2} rdrd\theta$$
$$= \int_0^{2\pi}\int_0^2 \frac{r^5}{2}drd\theta = \int_0^{2\pi}\left[\frac{r^6}{12}\right]_0^2 d\theta = \int_0^{2\pi}\frac{16}{3}d\theta = \frac{32}{3}\pi.$$

The value of M is

$$M = \int_0^{2\pi}\int_0^2\int_0^{r^2} dz\, r\, dr d\theta = \int_0^{2\pi}\int_0^2 \Big[z\Big]_0^{r^2} r dr d\theta$$
$$= \int_0^{2\pi}\int_0^2 r^3 dr d\theta = \int_0^{2\pi}\left[\frac{r^4}{4}\right]_0^2 d\theta = \int_0^{2\pi} 4 d\theta = 8\pi.$$

Therefore, $\bar{=}\frac{M_{xy}}{M} = \frac{32\pi}{3}\frac{1}{8\pi} = \frac{4}{3}$.
and the centroid is $(0, 0, 4/3)$. Notice that the centroid lies outside the solid.

Q52

$R = \{(x, y, z)| -2 \le x \le 2, -\sqrt{4-x^2} \le y \le \sqrt{4-x^2}, 0 \le z \le \sqrt{4-(x^2+y^2)}\}$
$= \{(\rho, \phi, \theta)|\, 0 \le \rho \le 2, 0 \le \phi \le \frac{\pi}{2}, 0 \le \theta \le 2\pi\}$.

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_0^{4-(x^2+y^2)} z^2\sqrt{x^2+y^2+z^2}dzdydx = \int_0^{2\pi}\int_0^{\pi/2}\int_0^2 (\rho^3\cos^2\phi)(\rho^2\sin\phi)d\rho d\phi d\theta$$
$$= \int_0^{2\pi}\int_0^{\pi/2}\left[\frac{1}{6}\rho^6\right]_0^2 d\phi d\theta = \frac{32}{3}\int_0^{2\pi}\int_0^{\pi/2}\cos^2\phi\sin\phi d\phi d\theta$$
$$= \frac{32}{3}\int_0^{2\pi}\left[\frac{-1}{3}\cos^3\phi\right]_0^{\pi/2} d\theta = \frac{32}{9}\int_0^{2\pi} d\theta = \frac{64\pi}{9}.$$

Q53

$x = u + \frac{v}{2}$ and $y = v \implies 2x - y = (2u+v) - v = 2u$ and $\frac{\partial(x,y)}{\partial(u,v)} = J(u,v) = \begin{vmatrix} 1 & 1/2 \\ 0 & 1 \end{vmatrix} = 1$;
next, $u = x - \frac{v}{2} = x - \frac{y}{2}$ and $v = y$, so the boundaries of the region of integration R in the xy-plane are transformed to the boundaries of G:

xy-equations for the boundary of R	Corresponding uv-equations for the boundary of G	Simplified uv-equations
x = y/2	u + v/2 = v/2	u = 0
x = y/2+2	u + v/2 = v/2 + 2	u = 2
y = 0	v = 0	v = 0
y = 2	v = 2	v = 2

$$\implies \int_0^2\int_{y/2}^{y/2+y} y^3(2x-y)e^{(2x-y)^2}dxdy = \int_0^2\int_0^2 v^3(2u)e^{4u^2}dudv = \int_0^2 v^3\left[\frac{1}{4}e^{4u^2}\right]_0^2 dv$$
$$= \frac{1}{4}\int_0^2 v^3(e^{16}-1)dv = \frac{1}{4}(e^{16}-1)\left[\frac{v^4}{4}\right]_0^2 = e^{16}-1.$$

Q54

Substitute $x = \rho\sin\phi\cos\theta, y = \rho\sin\phi\sin\theta, z = \rho\cos\phi$ into $z = \sqrt{3x^2+3y^2}$,
then $\rho\cos\phi = \sqrt{3(\rho\sin\phi\cos\theta)^2 + 3(\rho\sin\phi\sin\theta)^2} = \sqrt{3}\rho\sin\phi \implies \tan\phi = \frac{1}{\sqrt{3}} \implies \phi = \frac{\pi}{6}$.
$S = \{(\rho\phi, \theta)\,|0 \le \rho \le 4, 0 \le \phi \le \frac{\pi}{6}.0 \le \theta \le 2\pi\}$.

$$V = \int_0^{2\pi}\int_0^{\pi/6}\int_0^4 \rho^2\sin\phi d\rho d\phi d\theta = (2\pi)(1-\frac{\sqrt{3}}{2})(\frac{64}{3}) = \frac{64}{3}\pi(2-\sqrt{3}) = 17.958.$$

Q55

Note that $\sigma = \rho, \alpha = \arctan(1/2)$. On the base, $\rho\cos\phi = 2$.

The mass is $M = \int\int\int \rho dV = \int_0^{2\pi}\int_0^{\arctan(1/2)}\int_0^{2\sec\phi} \rho^3 \sin\phi d\rho d\phi d\theta = \dfrac{\pi}{3}(5\sqrt{5}-8)$.

Q56

Let $\widetilde{x} = x - 2$.

$\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} \int_1^{2\cos(\pi-\theta)} (r\cos\theta + 2) r dr d\theta = \frac{3\sqrt{3}}{4}$

Q57

$V = 4\int_0^{\frac{\pi}{2}} \int_0^{3\sin\theta} \int_0^{\sqrt{9-r^2}} dz r dr d\theta$

$= 4\int_0^{\frac{\pi}{2}} \int_0^{3\sin\theta} r\sqrt{9-r^2} dr d\theta$

$= 4\int_0^{\frac{\pi}{2}} [-\frac{1}{3}(9-r^2)^{\frac{3}{2}}]_0^{3\sin\theta} d\theta$

$= 4\int_0^{\frac{\pi}{2}} [-\frac{1}{3}(9-9\sin^2\theta)^{\frac{3}{2}} + \frac{1}{3}9^{\frac{3}{2}}] d\theta$

$= 36\int_0^{\frac{\pi}{2}} (1-(1-\sin^2\theta)^{\frac{3}{2}}) d\theta$

$= 36\int_0^{\frac{\pi}{2}} (1-\cos^3) d\theta$

$= 36\int_0^{\frac{\pi}{2}} (1-\cos\theta + \sin^2\theta\cos\theta) d\theta$

$= 36[\theta - \sin\theta + \frac{\sin^3\theta}{3}]_0^{\frac{\pi}{2}}$

$= 18\pi - 24$

Q58

$V = \int_0^R \int_0^{\sqrt{R^2-x^2}} (a-x-y) dy dx$

$= \int_0^R \int_0^{\frac{\pi}{2}} (a - \cos\theta\rho - \rho\sin\theta)\rho d\theta d\rho$

$= \int_0^R [a\rho\theta - \rho^2\sin\theta + \rho^2\cos\theta]_0^{\frac{\pi}{2}} d\rho$

$= \int_0^R (\frac{\pi}{2}a\rho - 2\rho^2) d\rho$

$= [\frac{\pi}{4}a\rho^2 - \frac{2}{3}\rho^3]_0^R$

$= \frac{\pi}{4}aR^2 - \frac{2}{3}R^3$

Q59

$= \int_0^1 \frac{1}{\sqrt{x}} \int_1^x \frac{\ln(1+t)}{t} dt dx$

$= -\int_0^1 \frac{\ln(1+t)}{t} dt \int_0^t \frac{1}{\sqrt{x}} dx$

$= -2\int_0^1 \frac{\ln(1+t)}{\sqrt{t}} dt \quad \text{let } t = u^2 \quad \frac{dt}{du} = 2u$

$= -2\int_0^1 \frac{\ln(1+u^2)}{u} 2u du$

$= -4\int_0^1 \ln(1+u^2) du$

$= -4u\ln(1+u^2)|_0^1 + 8\int_0^1 \frac{u^2}{1+u^2} du$

$= -4\ln 2 + 8\int_0^1 \frac{u^2+1-1}{1+u^2} du$

$= -4\ln 2 + 8 - 8\int_0^1 \frac{1}{1+u^2} du$

$= -4\ln 2 + 8 - 8\arctan u|_0^1$

$= -4\ln 2 + 8 - 8(\frac{\pi}{4} - 0)$

$= -4\ln 2 + 8 - 2\pi$

Q60

$f_x = \cos x - \cos(x+y) = 0, f_y = \cos y - \cos(x+y) = 0$

$\Rightarrow \cos x = \cos y = \cos(x+y)$

When $x > 0, y > 0, 0 < x+y < 2\pi \Rightarrow x = y = \frac{2}{3}\pi$

On the boundary $x = 0$ or $y = 0$ or $x + y = 2\pi, f(x,y) = 0$

Thus, $f_{max} = f(\frac{2}{3}\pi, \frac{2}{3}\pi) = \frac{3\sqrt{3}}{2}, f_{min} = 0.$

Q61

Let $u = \dfrac{y^2}{x}, v = \dfrac{y}{x} \quad x = \dfrac{u}{v^2}, y = \dfrac{u}{v}$

$\dfrac{\partial(x,y)}{\partial(u,v)} = \dfrac{u}{v^4}$

Thus, $S = \displaystyle\int_n^m udu \int_q^p \frac{dv}{v^4} = \frac{1}{6}(m^2 - n^2)(\frac{1}{q^3} - \frac{1}{p^3})$

Q62

Let $\tilde{y} = y - 2, \tilde{x} = r\cos\theta, \tilde{y} = r\sin\theta$

$\displaystyle\int_{\frac{7}{6}\pi}^{\frac{11}{6}\pi} \int_1^{2\cos(\frac{3}{2}\pi - \theta)} (r\sin\theta + 2) r dr d\theta = \int_{\frac{7}{6}\pi}^{\frac{11}{6}\pi} -1 - \frac{\sin\theta}{3} + 4\sin^2\theta - \frac{8\sin^4\theta}{3} d\theta = \frac{3\sqrt{3}}{4}$

Q63

$V = 4\int_0^{\frac{\pi}{2}} d\theta \int_0^{R\cos\theta} \sqrt{R^2 - r^2} r dr$

$= \frac{4}{3}R^3 \int_0^{\frac{\pi}{2}} (1 - \sin^3\theta) d\theta$

$= \frac{6\pi - 8}{9} R^3$

Q64

$x^2 + y^2 < 1$, find critical values

$x^2 + y^2 = 1$, Lagrange multiplies

① critical values $\quad T = x^2 + y^2 - x - y + xy$

$T_x = 2x - 1 + y = 0, T_y = 2y - 1 + x = 0$

$\Rightarrow (x,y) = (\frac{1}{3}, \frac{1}{3})$

② Lagrange multiplies $\begin{cases} 2x - 1 + y = 2x\lambda & \Rightarrow \lambda = \frac{2x-1+y}{2x} (x \neq 0) \\ 2y - 1 + x = 2y\lambda & \Rightarrow \lambda = \frac{-1+x+2y}{2y} (y \neq 0) \\ x^2 + y^2 = 1 \end{cases}$

If $x = 0, y = 1 \quad$ If $y = 0, x = 1 \Rightarrow (0,1), (1,0)$

If $x \neq 0, y \neq 0, \frac{2x-1+y}{2x} = \frac{2y-1+x}{2y} \Rightarrow x = y$ or $x + y - 1 = 0$

When $x = y, x^2 + y^2 = 1 \Rightarrow x = y = \pm\frac{\sqrt{2}}{2} \Rightarrow (\pm\frac{\sqrt{2}}{2}, \pm\frac{\sqrt{2}}{2})$

$x = 1 - y, x^2 + y^2 = 1 \Rightarrow x = 0, y = 1$ (delete)

$T(0,1) = 0, T(1,0) = 0, T(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}) = \frac{3}{2} - \sqrt{2}, T(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}) = \frac{3}{2} + \sqrt{2}, T(\frac{1}{3}, \frac{1}{3}) = -\frac{1}{3}$

Thus, hottest point $(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}), T = \frac{3}{2} + \sqrt{2}$, coldest point $(\frac{1}{3}, \frac{1}{3}), T = -\frac{1}{3}$.

Chapter 3

Q65

(a) $\begin{vmatrix} 1 & h & 1 \\ 0 & 1 & 2h \\ 0 & -h & 3h+1 \end{vmatrix} = \begin{vmatrix} 1 & 2h \\ -h & 3h+1 \end{vmatrix} = 2h^2 + 3h + 1 = (2h+1)(h+1) \neq 0 \implies h \neq -1/2, h \neq -1$

Thus, $h \in \mathbb{R} \setminus \{-1/2, -1\}$.

(b) Note that $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & k \\ 1 & 4 & k^2 \end{vmatrix} = k^2 - 3k + 2 = (k-1)(k-2) \neq 0$

Hence, $k \in \mathbb{R} \setminus \{1, 2\}$

(c) i. $2^n x$

ii. y

iii. 2p

Q66

a. F, any reasonable counterexample is acceptable, e.g. system of homogeneous equations.

b. F e.g. $0x + 0y = 1$ has no solution.

c. F, e.g. $\begin{pmatrix} 1 \\ 1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

d. T. The trivial solution is always the solution to homogeneous system.

Q67

a. Note that $\left|A - \lambda I_3\right| = \begin{vmatrix} 5-\lambda & 4 & 2 \\ 4 & 5-\lambda & 2 \\ 2 & 2 & 2-\lambda \end{vmatrix} = -(\lambda - 10)(\lambda - 1)^2$ Thus, eigenvalues are 1 (double) and 10.

For $\lambda = 1$,

$$\begin{pmatrix} 4 & 4 & 2 \\ 4 & 4 & 2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \implies x = \begin{pmatrix} -t - s/2 \\ t \\ s \end{pmatrix}$$

Thus, the eigenvectors are $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$.

For $\lambda = 10$,

$$\begin{pmatrix} -5 & 4 & 2 \\ 4 & -5 & 2 \\ 2 & 2 & -8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \implies x = \begin{pmatrix} 1 \\ 1 \\ 1/2 \end{pmatrix} t$$

Thus, the eigenvector is $\begin{pmatrix} 1 \\ 1 \\ 1/2 \end{pmatrix}$.

b. Note that $P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & -2 & 1/2 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10 \end{pmatrix}$.

c. $A^{2018} = (PDP^{-1})^{2018} = PD^{2018}P^{-1}$

$$= \frac{1}{9} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & -2 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10^{2018} \end{pmatrix} \begin{pmatrix} 4 & -5 & 2 \\ 1 & 1 & -4 \\ 4 & 4 & 2 \end{pmatrix}$$

$$= \frac{1}{9} \begin{pmatrix} 1 & 1 & 10^{2018} \\ -1 & 0 & 10^{2018} \\ 0 & -2 & 5 \times 10^{2017} \end{pmatrix} \begin{pmatrix} 4 & -5 & 2 \\ 1 & 1 & -4 \\ 4 & 4 & 2 \end{pmatrix}$$

$\implies$ the (3,3) entry is $\frac{1}{9}(8 + 10^{2018})$

Q68

The answer is $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Note that $\begin{pmatrix} 1 & -2 & 3 \\ 2 & -3 & 2 \\ 1 & 2 & -5 \end{pmatrix}^{-1} = \begin{pmatrix} 11/8 & -1/2 & 5/8 \\ 3/2 & -1 & 1/2 \\ 7/8 & -1/2 & 1/8 \end{pmatrix}$

Q69

Use Gaussian elimination on the augmented matrix.

$$\begin{bmatrix} (\lambda+2) & -2 & 3 & 0 \\ -2 & (\lambda-1) & 6 & 0 \\ 1 & 2 & \lambda & 0 \end{bmatrix} \implies \begin{bmatrix} 1 & 2 & \lambda & 0 \\ 0 & \lambda+3 & 6+2\lambda & 0 \\ 0 & -2\lambda-6 & -\lambda^2-2\lambda+3 & 0 \end{bmatrix} \implies \begin{bmatrix} 1 & 2 & \lambda & 0 \\ 0 & \lambda+3 & 6+2\lambda & 0 \\ 0 & 0 & (\lambda^2-2\lambda-15) & 0 \end{bmatrix}$$

So, you need $\lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3) = 0 \implies \lambda = 5$ or $\lambda = -3$.

Q70

a. i. No, a homogeneous system always has the trivial solution.

ii. $\beta = 2$.

Note that $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ -1 & 1 & \beta \end{pmatrix} \xrightarrow[R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 4 & 1+\beta \end{pmatrix} \xrightarrow{-3R_2+R_3\to R_3} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & \beta-2 \end{pmatrix}$.

b. i. $a = 5$ and $b = 4$.

ii. $a = 5$ and $b \neq 4$

iii. $a \neq 5$.

Note that $\begin{pmatrix} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{pmatrix} \xrightarrow[R_1+R_3\to R_3]{-R_1+R_2\to R_2} \begin{pmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & a-4 & b-2 \end{pmatrix} \xrightarrow{-2R_2+R_3\to R_3} \begin{pmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & b-4 \end{pmatrix}$.

Q71

a. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ which has eigenvalues $\lambda_1 = 5$ with eigenvector $(1, 2, -1)^T$, $\lambda_2 = -3$ (repeated), with eigenvectors $(1, 0, 1/3)^T, (1, -1/2, 0)^T$.

b. $P = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & -1/2 \\ -1 & 1/3 & 0 \end{pmatrix}, D = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}, P^{-1} = \begin{pmatrix} 1/8 & 1/4 & -3/8 \\ 3/8 & 3/4 & 15/8 \\ 1/2 & -1 & -3/2 \end{pmatrix}$

c. $\begin{pmatrix} \frac{p+7q}{8} & \frac{p-q}{4} & \frac{-3p+3q}{8} \\ \frac{p-q}{4} & \frac{p+q}{2} & \frac{-3p+3q}{4} \\ \frac{-p+q}{8} & \frac{-p+q}{4} & \frac{3p+5q}{8} \end{pmatrix}$, where $p = 5^{2017}, q = (-3)^{2017}$.

Q72

a. 3

b. If A is nilpotent, then $(A^k)^T = (A^T)^k = O$.

But,$(A^T)^{k-1} = (A^{k-1})^T \neq O$, which shows that A^T is nilpotent with the same index.

c. No. If A is nilpotent and invertible, then $A^k = O$ for some k and $A^{k-1} \neq O$. So, $A^{-1}A = I \implies O = A^{-1}A^k = (A^{-1}A)A^{k-1} = IA^{k-1} \neq O$ which is impossible.

d. Let A be nilpotent of index k. Then

$(I - A)(A^{k-1} + A^{k-2} + \cdots + a^2 + a + I) = I - A^k = I$

which shows that $(A^{k-1} + A^{k-2} + \cdots + a^2 + a + I)$ is the inverse pf $I - A$.

Q73 a). The characteristic polynomial of A is $\left|\lambda I - A\right| = \begin{vmatrix} \lambda-1 & 0 & 0 & 0 \\ 0 & \lambda-1 & -5 & 10 \\ -1 & 0 & \lambda-2 & 0 \\ -1 & 0 & 0 & \lambda-3 \end{vmatrix} = (\lambda-1)^2(\lambda-2)(\lambda-3)$.

So, the characteristic equation is $(\lambda-1)^2(\lambda-2)(\lambda-3) = 0$ and the eigenvalues are $\lambda_1 = 1$ with eigen vector $(0,1,0,0)^T, (1,0,-1,-1/2)^T$, $\lambda_2 = 2$, with eigenvector $(0,1,15,0)^T$ and $\lambda_3 = 3$ with eigenvector $(0,1,0,-1/5)^T$.

b).

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & -1 & 1/5 & 0 \\ 0 & -1/2 & 0-1/5 & \end{pmatrix}, P^{-1} = \begin{pmatrix} -5/2 & 1 & -5 & 5 \\ 1 & 0 & 0 & 0 \\ 5 & 0 & 5 & 0 \\ -5/2 & 0 & 0 & -5/2 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

c). $A^{2016} = PD^{2016}P^{-1} = \begin{pmatrix} -5/2 & 1 & -5 & 5 \\ 1 & 0 & 0 & 0 \\ 5 & 0 & 5 & 0 \\ -5/2 & 0 & 0 & -5/2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & k & 0 \\ 0 & 0 & 0 & m \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & -1 & 1/5 & 0 \\ 0 & -1/2 & 0-1/5 & \end{pmatrix}$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{10k-5m-5}{2} & 1 & 5k-5 & -5m+5 \\ k-1 & 0,k & k & 0 \\ \frac{m-1}{2} & 0 & 0 & m \end{pmatrix}, \text{ where } k = 2^{2016} \text{ and } m = 3^{2016}.$$

Q74

a).

$$x = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 8 & 11 \\ 1 & 5 & 7 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & -1 \\ -5 & 2 & -1 \\ 7/2 & -3/2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 6 \\ -4 \end{bmatrix}.$$

b).

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 8 & 11 \\ 1 & 5 & 7 \end{vmatrix} = 2, \Delta_1 \begin{vmatrix} -1 & 2 & 3 \\ 1 & 8 & 11 \\ 1 & 5 & 7 \end{vmatrix} = -2, \Delta_2 = \begin{vmatrix} 1 & -1 & 3 \\ 3 & 1 & 11 \\ 1 & 1 & 7 \end{vmatrix} = 12,$$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 8 & 1 \\ 1 & 5 & 1 \end{vmatrix} = -8.$$

Therefore, $x_1 = \dfrac{\Delta_1}{\Delta} = -1, x_2 = \dfrac{\Delta_2}{\Delta} = 6, x_3 = \dfrac{\Delta_3}{\Delta} = -4.$

Q75 The solution is $(1, -1, 2)$.

Q76

Note that $[A \quad \mathbf{b}] = \begin{bmatrix} 1 & 3 & -2 & 5 & b_1 \\ 5 & 8 & -5 & 12 & b_2 \\ 3 & 16 & -11 & 28 & b_3 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & 3 & -2 & 5 & b_1 \\ 0 & -7 & 5 & -13 & b_2 - 5b_1 \\ 0 & 7 & -5 & 13 & b_3 - 3b_1 \end{bmatrix}$

$$\Longrightarrow \begin{bmatrix} 1 & 3 & -2 & 5 & b_1 \\ 0 & 1 & -5/7 & 13/7 & (5b_1 - b_2)/7 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 8b_1 \end{bmatrix}.$$

Therefore, the system is consistent if and only if $b_3 + b_2 - 8b_1 = 0$.

b).

When $\mathbf{b} = [1 \quad 4 \quad 4]^T$,

$$\begin{cases} x_1 + 3x_2 - 2x_3 + 5x_4 = 1 \\ x_2 - 5/7x_3 + 13/7x_4 = 1/7 \end{cases}$$

Therefore, solutions are $x = [\frac{-t}{7} + \frac{4s}{7} + \frac{4}{7} \quad \frac{5t}{7} - \frac{13s}{7} + \frac{1}{7} \quad t \quad s]^T$

Q77

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 2 & 0 & 4 \\ 0 & 2-\lambda & -1 & 0 \\ 0 & 0 & 3-\lambda & 3 \\ 0 & 4 & 0 & 7-\lambda \end{vmatrix} = \lambda^4 - 16\lambda^3 + 89\lambda^2 - 194\lambda + 120$$

$= (\lambda - 1)(\lambda - 4)(\lambda - 5)(\lambda - 6) = 0.$

Thus, the fourth eigenvalue of A is $\lambda = 5$.

When $\lambda = 5, (A - \lambda I)\mathbf{v} = \mathbf{0} \implies \begin{bmatrix} -1 & 2 & 0 & 4 & 0 \\ 0 & -3 & -1 & 0 & 0 \\ 0 & 0 & -2 & 3 & 0 \\ 0 & 4 & 0 & 2 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -3 & 0 \\ 0 & 1 & 0 & 1/2 & 0 \\ 0 & 0 & 10 & -3/2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

Therefore, $\mathbf{v} = \begin{bmatrix} 3 \\ -1/2 \\ 3/2 \\ 1 \end{bmatrix} s$, where $s \in \mathbb{R}$, and, the eigenvector associated with $\lambda = 5$ is $\begin{bmatrix} 1 \\ -1/6 \\ 3/6 \\ 1/3 \end{bmatrix}$.

Q78

a). $\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_{n-1} \\ a_{n-2} \end{bmatrix} = \begin{bmatrix} a_{n-1} + 2a_{n-2} \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} a_n \\ a_{n-1} \end{bmatrix}$, for $n \geq 3. A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$

b). $A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ has eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$, with eigenvector $v^1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $v^2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ respectively.

Therefore, one obtain $P = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$.

c).

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}^{-1}.$$

$$A^{2016} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^{2016} & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2^{2017} + 1 & 2^{2017} - 2 \\ 2^{2016} - 1 & 2^{2016} + 2 \end{bmatrix}$$

Q79

a). Consider

$$0 = \det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = \begin{vmatrix} -\lambda & 1+\lambda & 1 \\ 1 & -\lambda - 1 & 1 \\ 1 & 0 & -\lambda \end{vmatrix}$$

$$= (1+\lambda)\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -\lambda \end{vmatrix} = (1+\lambda)\begin{vmatrix} -\lambda & 1 & 1 \\ 1-\lambda & 0 & 2 \\ 1 & 0 & -\lambda \end{vmatrix} = -(1+\lambda)(\lambda^2-\lambda-2) = -(1+\lambda)^2(\lambda-2).$$

Thus, the eigenvalues are 2 and -1 (repeated).

When $\lambda = 2$, we have

$$\begin{bmatrix} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} 0 & -3 & 3 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 3 & -3 & 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

i.e. $x_1 = x_3$ and $x_2 = x_3$. The corresponding eigenvector is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

When $\lambda = -1$, we have

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

i.e. $x_1 + x_2 + x_3 = 0$. The corresponding eigenvector is $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

b).

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \text{ and } D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

c).

$$P^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}, \ A = PDP^{-1}.$$

Now, $A^{2015} = PD^{2015}P^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}\begin{bmatrix} 2^{2015} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

$$= \frac{1}{3}\begin{bmatrix} 2^{2015}-2 & 2^{2015}+1 & 2^{2015}+1 \\ 2^{2015}+1 & 2^{2015}-2 & 2^{2015}+1 \\ 2^{2015}+1 & 2^{2015}+1 & 2^{2015}-1 \end{bmatrix}.$$

Q80 $x = (2.3.4)^T$.

a). $\begin{pmatrix} 2 & -2 & 3 \\ 3 & -3 & 2 \\ 4 & 2 & -5 \end{pmatrix}^{-1} = \begin{pmatrix} 11/30 & -2/15 & 1/6 \\ 23/30 & -11/15 & 1/6 \\ 3/5 & -2/5 & 0 \end{pmatrix}$.

Then, $x = \begin{pmatrix} 11/30 & -2/15 & 1/6 \\ 23/30 & -11/15 & 1/6 \\ 3/5 & -2/5 & 0 \end{pmatrix}\begin{pmatrix} 10 \\ 5 \\ -6 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

b). Note that $\Delta = \begin{vmatrix} 2 & -2 & 3 \\ 3 & -3 & 2 \\ 4 & 2 & -5 \end{vmatrix} = 30$. $\Delta_1 = \begin{vmatrix} 10 & -2 & 3 \\ 5 & -3 & 2 \\ -6 & 2 & -5 \end{vmatrix} = 60$.

$\Delta_2 = \begin{vmatrix} 2 & 10 & 3 \\ 3 & 5 & 2 \\ 4 & -6 & -5 \end{vmatrix} = 90$. $\Delta_3 = \begin{vmatrix} 2 & -210 & \\ 3 & -3 & 5 \\ 4 & 2 & -6 \end{vmatrix} = 120$.

Hence, $x = (\frac{\Delta_1}{\Delta}, \frac{\Delta_2}{\Delta}, \frac{\Delta_3}{\Delta})^T = (2, 3, 4)^T$

Q81

The eigenvalues of Aa are the solutions of $|\lambda I - A| = \begin{vmatrix} \lambda - 2 & 1 & -1 \\ 2 & \lambda - 3 & 2 \\ 1 & -1 & \lambda \end{vmatrix} = (\lambda - 1)^2(\lambda - 3) = 0$

Therefore, the eigenvalues are $\lambda_1 = 1$ (repeated) and $\lambda_2 = 3$. The corresponding eigenvectors are solutions of $(\lambda I - A)\mathbf{x} = \mathbf{0}$.

So $(1, 0, -1)$ and $(1, 1, 0)$ are eigenvectors corresponding to $\lambda_1 = 1$, while $(1, -2, -1)$ corresponding to $\lambda_2 = 3$. Now form P from these eigenvectors and note that

$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ -1 & 0 & -1 \end{bmatrix}$ and $P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

Note that $P^{-1} = \begin{bmatrix} -1/2 & 1/2 & -3/2 \\ 1 & 0 & 1 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}$.

Hence, $A^{2017} = \begin{bmatrix} \frac{k+1}{2} & \frac{-k+1}{2} & \frac{k-1}{2} \\ -k+1 & k & -k+1 \\ \frac{-k+1}{2} & \frac{k-1}{2} & \frac{-k+3}{2} \end{bmatrix}$, where $k = 3^{2017}$.

Q82

Find $A^{-1} = \begin{bmatrix} 11 & 2 & -8 \\ 4 & 1 & -3 \\ -8 & -1 & 6 \end{bmatrix}$, and multiply each coded row matrix on the right by A^{-1} to find the associated uncoded row matrix.

Note that $[112 \quad -140 \quad 83]A^{-1} = [112 \quad -120 \quad 83] \begin{bmatrix} 11 & 2 & -8 \\ 4 & 1 & -3 \\ -8 & -1 & 6 \end{bmatrix} = [8 \quad 1 \quad 22] \implies$ H,A,V

$[19 \quad -25 \quad 13]A^{-1} \implies [5 \quad 0 \quad 1] \implies$ E, _,A

$[72 \quad -76 \quad 61]A^{-1} = [0 \quad 7 \quad 18] \implies$ _ G R.

$[959 \quad -118 \quad 71]A^{-1} = [5 \quad 1 \quad 20] \implies$ E A T.

$[20 \quad 21 \quad 38]A^{-1} = [0 \quad 23 \quad 5] \implies$ _ W E.

$[35 \quad -23 \quad 36]A^{-1} = [5 \quad 11 \quad 5] \implies$ E K E.

$[42 \quad -48 \quad 32]A^{-1} = [14 \quad 4 \quad 0] \implies$ N D _.

Thus, the message is HAVE_A_GREAT_WEEKEND_ .

Q83

a). $\begin{vmatrix} \lambda+1 & 2 & 2 \\ -1 & \lambda-2 & -1 \\ 1 & 1 & \lambda \end{vmatrix} = 0 \implies \lambda^3 - \lambda^2 + 1 = 0 \implies (\lambda-1)^2(\lambda+1)$

Hence, $\lambda_1 = 1 = \lambda_2$ and $\lambda_3 = -1$.

For $\lambda_1, \lambda_2 = 1$,

the corresponding eigenvectors are $v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$.

For $\lambda_3 = -1$, the corresponding eigenvector is $v_3 = \begin{bmatrix} 1 \\ -1/2 \\ 1/2 \end{bmatrix}$

b). $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1/2 \\ -1 & -1 & 1/2 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

c).

$P^{-1} = \begin{bmatrix} 0 & -1 & -1 \\ 1/2 & 3/2 & 1/2 \\ 1 & 1 & 1 \end{bmatrix}$

d).

$A^{2018} = (PDP^{-1})^{2018} = PP^{-1} = I_3$.

Q84 a). i.No

ii.yes

b). $P = \begin{pmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{pmatrix}, x = \begin{pmatrix} 15,000 \\ 20,000 \\ 65,000 \end{pmatrix}$.

Hence, $Px = \begin{pmatrix} 23,250 \\ 28,750 \\ 48,000 \end{pmatrix}$

$A_n \quad R_1 + R_2 + R_3 + \cdots + R_n \to R_1 \quad \Rightarrow R_1 = (0, 0, \cdots, 0)$

Thus, $|A_n| = 0$.

$B_n - 1 \times R_1 + R_2 \to R_2$

$-1 \times R_1 + R_3 \to R_3$

$\cdots$

$-1 \times R_1 + R_n \to R_n$

$$\Rightarrow \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & -x & 0 & \cdots & 0 \\ 0 & 0 & 1-x & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & n-2-x \end{pmatrix}$$

Thus, $|B_n| = (-x)(1-x)(2-x)\cdots(n-2-x)$

a)

$\begin{vmatrix} 2 & 1 & 5 \\ 4 & 3 & 1 \\ 3 & 2 & 3 \end{vmatrix} = 0$, thus linearly dependent.

b)

$$\begin{pmatrix} 2 & 1 & 5 \\ 4 & 3 & 1 \\ 3 & 2 & 3 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1 \to R_1} \begin{pmatrix} 1 & \frac{1}{2} & \frac{5}{2} \\ 4 & 3 & 1 \\ 3 & 2 & 3 \end{pmatrix} \xrightarrow[-3R_1+R_3 \to R_3]{-4R_1+R_2 \to R_2} \begin{pmatrix} 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & -9 \\ 0 & \frac{1}{2} & -\frac{9}{2} \end{pmatrix} \xrightarrow{-\frac{1}{2}R_2+R_3 \to R_3} \begin{pmatrix} 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & -9 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} -7x_3 \\ 9x_3 \\ x_3 \end{pmatrix}$$

Thus, $-7v_1 + 9v_2 + v_3 = 0$.

Q85

a)

$$|A - \lambda I| = \begin{vmatrix} 5-\lambda & 8 & 16 \\ 4 & 1-\lambda & 8 \\ -4 & -4 & -11-\lambda \end{vmatrix} = (\lambda - 1)(\lambda + 3)^2$$

Thus, $\lambda_1 = 1 \lambda_{2,3} = -3$

①$\lambda_1 = 1$,

$$\begin{pmatrix} 4 & 8 & 16 \\ 4 & 0 & 8 \\ -4 & -4 & -12 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 0$$

$$V_1 = \begin{pmatrix} v_1 \\ \frac{1}{2}v_1 \\ -\frac{1}{2}v_1 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} v_1$$

Thus, the corresponding eigenvector to λ_1 is $\begin{pmatrix} 1 \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$

② $\lambda_{2,3} = -3$

$$\begin{pmatrix} 8 & 8 & 16 \\ 4 & 4 & 8 \\ -4 & -4 & -8 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 0$$

$$V_2 = \begin{pmatrix} v_1 \\ v_2 \\ -\frac{1}{2}v_1 - \frac{1}{2}v_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -\frac{1}{2} \end{pmatrix} v_1 + \begin{pmatrix} 0 \\ 1 \\ -\frac{1}{2} \end{pmatrix} v_2$$

Thus, the corresponding eigenvectors to $\lambda_{2,3}$ are $\begin{pmatrix} 1 \\ 0 \\ -\frac{1}{2} \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ -\frac{1}{2} \end{pmatrix}$

b)

$$P = \begin{pmatrix} 1 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix} \quad P^{-1} = \begin{pmatrix} 2 & 2 & 4 \\ -1 & -2 & -4 \\ -1 & 0 & -2 \end{pmatrix}$$

c)

$$A = (PDP^{-1})^{2019} = PD^{2019}P^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & (-3)^{2019} & 0 \\ 0 & 0 & (-3)^{2019} \end{pmatrix} \begin{pmatrix} 2 & 2 & 4 \\ -1 & -2 & -4 \\ -1 & 0 & -2 \end{pmatrix}$$

Thus, the $(3,3)$ entry is $-3^{2020} - 2$.

Q86

a)

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$\lambda_1 = 1, \lambda_{2,3} = 2$

$$\lambda_1 = 1, \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} V_1 = 0 \Rightarrow V_1 = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$\lambda_{2,3} = 2, \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} V_{2,3} = 0 \Rightarrow V_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} V_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

b)

$$P = \begin{pmatrix} 1 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \\ -\frac{1}{2} & -1 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

c)

$$A = PDP^{-1} \quad A^{2019} = PD^{2019}P^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \\ -\frac{1}{2} & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^{2019} & 0 \\ 0 & 0 & 2^{2019} \end{pmatrix} \begin{pmatrix} 2 & 0 & 2 \\ -1 & 0 & -2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2^{2019} + 2 & 0 & -2^{2020} + 2 \\ 2^{2019} - 1 & 2^{2019} & 2^{2019} - 1 \\ 2^{2019} - 1 & 0 & 2^{2020} - 1 \end{pmatrix}$$

Thus, $(3,3)$-entry of A^{2019} is $2^{2020} - 1$.